

CSA0308-DATA STRUCTURES

C04- AT2- Coding Golf Competition

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1. QUICK SORT

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QUESTIONS**Q1**
Quick Sort
5 marks**Q2**
Bubble sort
5 marks**Q3**
Insertion Sort
5 marks**Q4**
Merge sort
5 marks

1/4 answered

Q1 Quick Sort 5 marks

Write a program to sort the array 50, 23, 9, 18, 61, 32 using Quick Sort.

Your Code**Input**
6
50 23 9 18 61 32**Output**
9 18 23 32 50 61

Run Save

2. BUBBLE SORT



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QUESTIONS

Q1
Quick Sort
5 marks

Q2
Bubble sort
5 marks

Q3
Insertion Sort
5 marks

Q4
Merge sort
5 marks

3/4 answered

Q2 Bubble sort5 marks

Write a program that implements the bubble sort

Your Code

```
1 #include <stdio.h>
2 int main() {int n,i,j,t; if(scanf("%d",&n)!=1) return 0; int a[n];
3 for(i=0;i<n;i++) scanf("%d",&a[i]);
4 for(i=0;i<n-1;i++) for(j=0;j<n-1-i;j++) if(a[j]>a[j+1]) t=a[j],a[j]=a[j+1],a[j+1]=t;
5 for(i=0;i<n;i++) printf("%d\tc",a[i],i==n-1?'\\n':' ');
6 }
```

Input

5
5 2 9 1 5

Output

1 2 5 9

Run

Save

3. INSERTION SORT

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QUESTIONS

Q1
Quick Sort
5 marks**Q2**
Bubble sort
5 marks**Q3**
Insertion Sort
5 marks**Q4**
Merge sort
5 marks

3/4 answered

Q3 Insertion Sort

5 marks

Write a program sort the elements 9, 5, 1, 4, 3 using Insertion Sort.

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int a[] = {9, 5, 1, 4, 3}, n = 5, i, j, k;
5     for (i = 1; i < n; i++) {
6         k = a[i];
7         for (j = i - 1; j >= 0 && a[j] > k; j--)
8             a[j + 1] = a[j];
9         a[j + 1] = k;
10    }
11    for (i = 0; i < n; i++) printf("%d ", a[i]);
12    return 0;
13 }
```

Input

9 5 1 4 3

Output

1 3 4 5 9

[▶ Run](#)[Save](#)

4. MERGE SORT

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QUESTIONS

Q1
Quick Sort
5 marks

Q2
Bubble sort
5 marks

Q3
Insertion Sort
5 marks

Q4
Merge sort
5 marks

4/4 answered

Q4 Merge sort 5 marks

Apply Merge Sort to the array 38, 27, 43, 3, 9, 82, 10.

Your Code

```
1 #include <stdio.h>
2
3 void s(int *a, int l, int r) {
4     if (l >= r) return;
5     int m = l + (r - l) / 2, i = l, j = m + 1, k = 0, t[r - l + 1];
6     s(a, l, m);
7     s(a, m + 1, r);
8     while (i <= m && j <= r) t[k++] = (a[i] <= a[j]) ? a[i++] : a[j++];
9     while (i <= m) t[k++] = a[i++];
10    while (j <= r) t[k++] = a[j++];
11    for (i = 0; i < k; i++) a[l + i] = t[i];
12 }
13
14 int main() {
15     int n, i;
16     if (scanf("%d", &n) != 1) return 0;
17     int a[n];
18     for (i = 0; i < n; i++) scanf("%d", &a[i]);
19     s(a, 0, n - 1);
20     for (i = 0; i < n; i++) printf("%d%c", a[i], i == n - 1 ? '\n' : ' ');
21     return 0;
22 }
```

Input

```
7
38 27 43 3 9 82 10
```

Output

```
3 9 10 27 38 43 82
```

Run Save